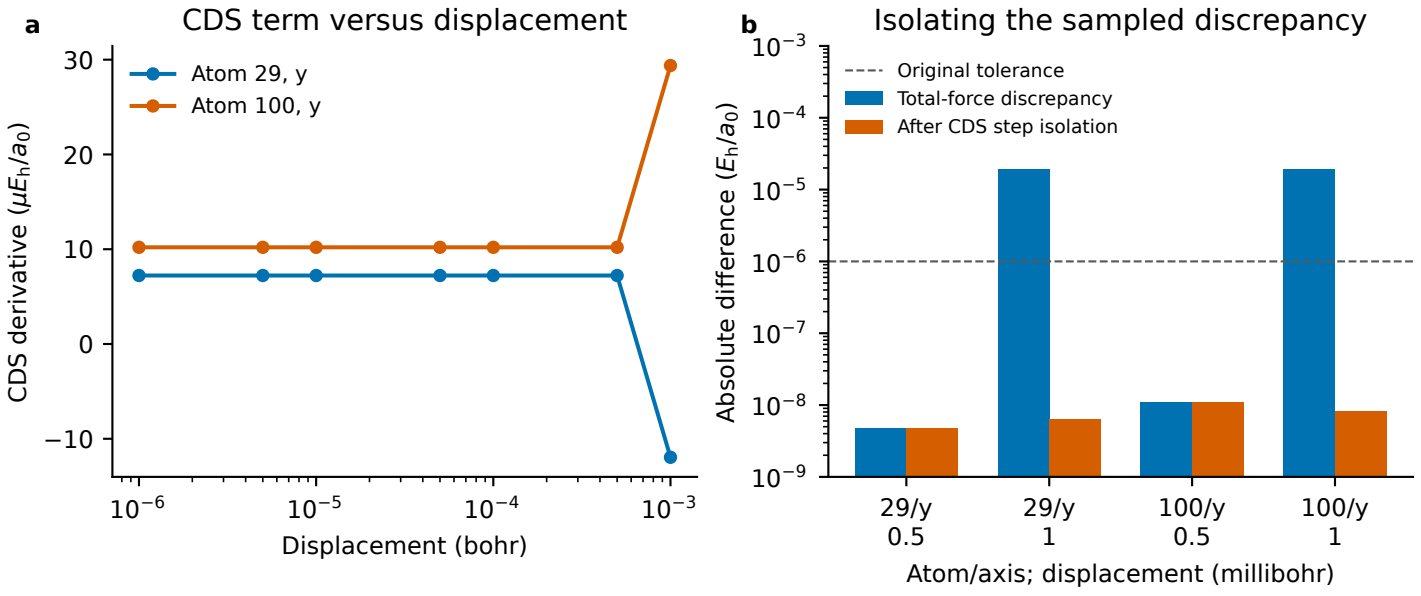
a
CDS term versus displacement
Atom 29, y
Atom 100, y
CDS derivative ($\mu E_h/a_0$)
30
20
10
0
−10
10^{-6}
10^{-5}
10^{-4}
10^{-3}
Displacement (bohr)
b
Isolating the sampled discrepancy
Original tolerance
Total-force discrepancy
After CDS step isolation
Absolute difference (E_h/a_0)
10^{-3}
10^{-4}
10^{-5}
10^{-6}
10^{-7}
10^{-8}
10^{-9}
29/y 0.5
29/y 1
100/y 0.5
100/y 1
Atom/axis; displacement (millibohr)